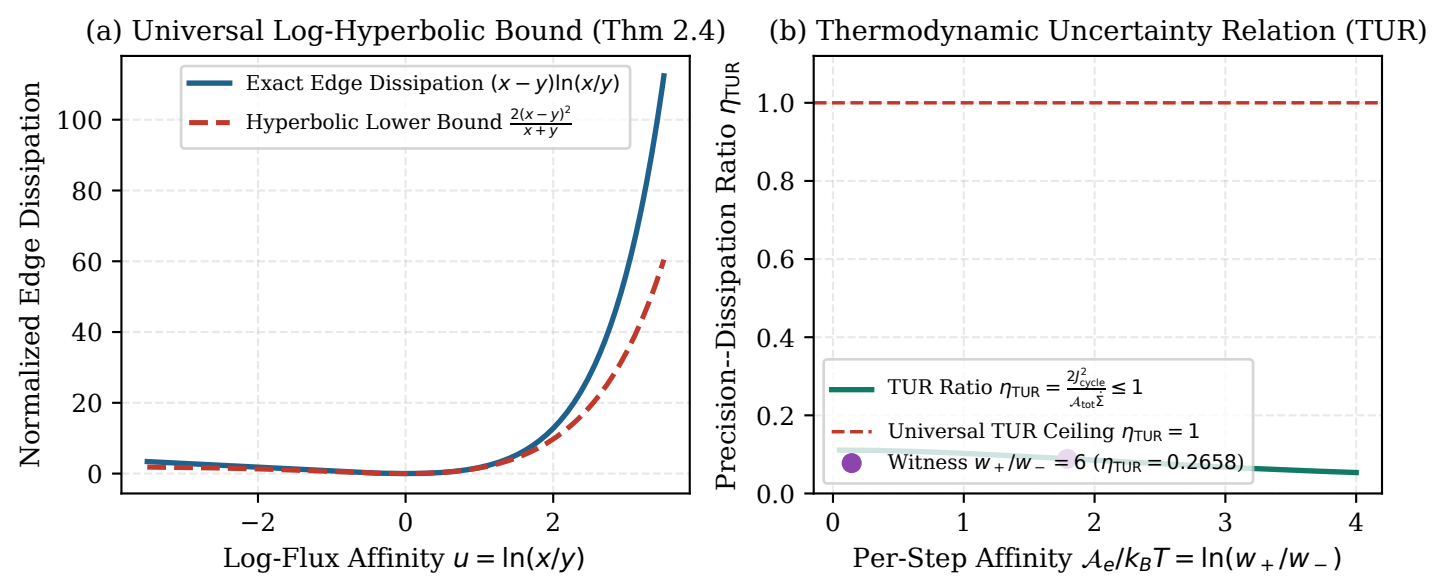
(a) Universal Log-Hyperbolic Bound (Thm 2.4)
Exact Edge Dissipation $(x-y)\ln(x/y)$
Hyperbolic Lower Bound $\frac{2(x-y)^2}{x+y}$
Normalized Edge Dissipation
Log-Flux Affinity $u=\ln(x/y)$
(b) Thermodynamic Uncertainty Relation (TUR)
TUR Ratio $\eta_{\mathrm{TUR}}=\frac{2J_{\mathrm{cycle}}^2}{\mathcal{A}_{\mathrm{tot}}\Sigma}\le 1$
Universal TUR Ceiling $\eta_{\mathrm{TUR}}=1$
Witness $w_+/w_-=6$ ($\eta_{\mathrm{TUR}}=0.2658$)
Precision--Dissipation Ratio η_{TUR}
Per-Step Affinity $\mathcal{A}_e/k_BT=\ln(w_+/w_-)$